At this point, let's not worry about the meaning of the symbols except to note that it is an equation that tells you how something changes with time. (The "something" is a representation of the state-vector of a particle.)

The iconic Schrödinger equation is a special case of a more general equation that we've already met in Eq. 4.9. It is part definition and part principle of quantum mechanics. As a principle, it says that the state-vector changes continuously with time, in a unitary way. As a definition, it defines the Hamiltonian, and therefore the observable called energy. Eq. 4.10,

$$\hbar \frac{\partial |\Psi\rangle}{\partial t} = -i\mathbf{H}|\Psi\rangle,$$

is sometimes called the *time-dependent* Schrödinger equation. Because the Hamiltonian operator **H** represents energy, the observable values of energy are just the eigenvalues of **H**. Let's call these eigenvalues $E_j$ and the corresponding eigenvectors $|E_j\rangle$. By definition, the relation between **H**, $E_j$, and $|E_j\rangle$ is the eigenvalue equation

$$\mathbf{H}|E_j\rangle = E_j|E_j\rangle. \qquad (4.28)$$

This is the *time-independent Schrödinger equation*, and it's used in two different ways.

If we work in a particular matrix basis, then the equation determines the eigenvectors of **H**. One puts in a particular value of the energy $E_j$ and looks for the ket-vector $|Ej\rangle$ that solves the equation.

It is also an equation that determines the eigenvalues $E_j$. If you put in an arbitrary value of $E_j$, in general there will not be a solution for the eigenvector. Let's take a very simple example: Suppose the Hamiltonian is the matrix $\frac{\hbar\omega}{2}\sigma_z$. Since $\sigma_z$ has only two eigenvalues, namely ±1, the Hamiltonian also has only two eigenvalues, $\pm\frac{\hbar\omega}{2}$. If you put any other value on the right-hand side of Eq. 4.28, there will not be a solution. Because the